

Case

A 54 yo man presents with 6 months: spontaneous chronic aching, stiffness and pain in the knuckles with some warmth and tenderness.

PMH: “liver inflammation”

Erectile dysfunction

Diabetes mellitus

SH: no tob/etoh; distant intranasal cocaine

FH: similar msk and other health problems in an older brother

What test is most likely to yield a diagnosis:

HLA-B27

HFE

HLA-DRW

IgG4 levels

angiotensin converting enzyme

B

R



R



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HLA-B27

HFE

HLA-DRW

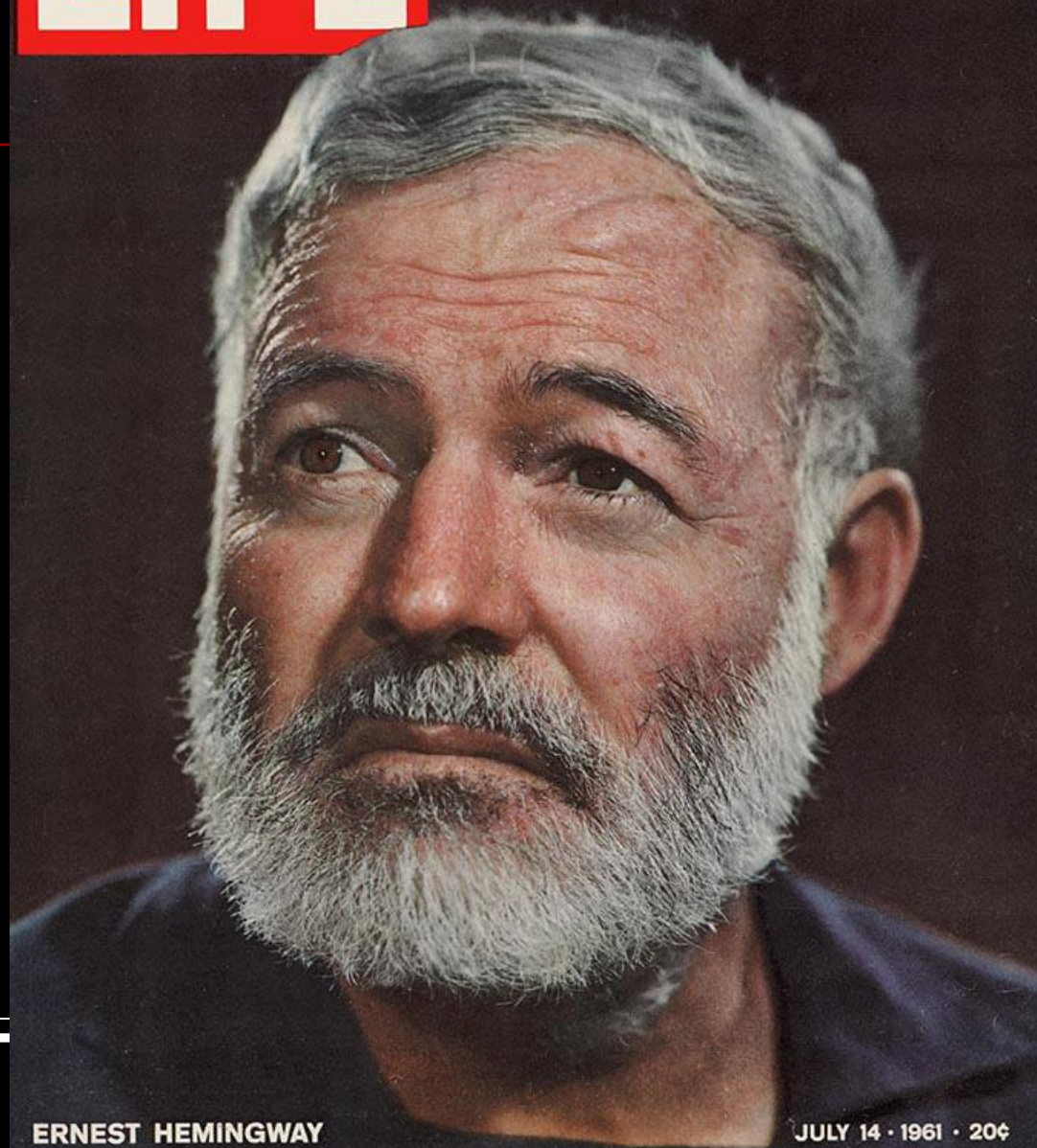
IgG4 levels

angiotensin converting enzyme

LIFE

THE CALL OF DANGER

In pictures, a great writer's
fascination with bravery and death.
With a memoir by Archibald MacLeish



ERNEST HEMINGWAY

JULY 14 • 1961 • 20¢

Calcium Pyrophosphate Disease (CPPD)

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CPPD

- An inflammatory response to the intraarticular deposition of calcium pyrophosphate dihydrate (CPPD) crystals
- Presentation can be similar to gout and was thus “Pseudogout” – 1962
- Epidemiology: aging, genetics, endocrinopathies
 - 3% of >60
 - 50% of >90
- Can also cause chronic synovitis (“pseudo-RA”); incidental on imaging
- Most common joints?
 - knee, wrist
 - shoulder, 2nd and 3rd MCP

CPPD XR findings

- Linear densities in cartilage:
 - medial and/or lateral menisci
 - wrists (triangular fibrocartilage)
- Hooked mcp osteophytes
- Isolated patellofemoral narrowing or wrist or glenohumeral “DJD”

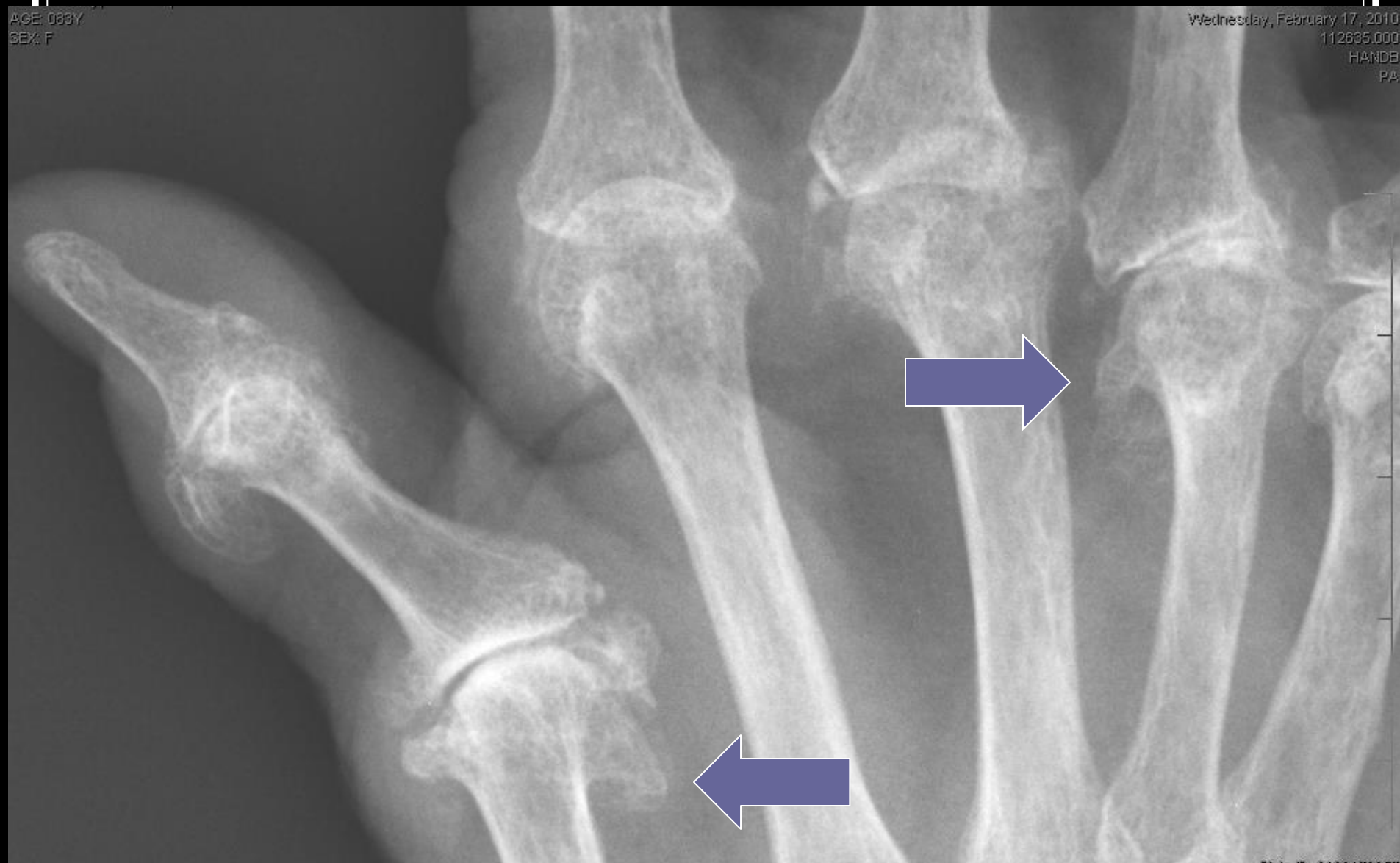
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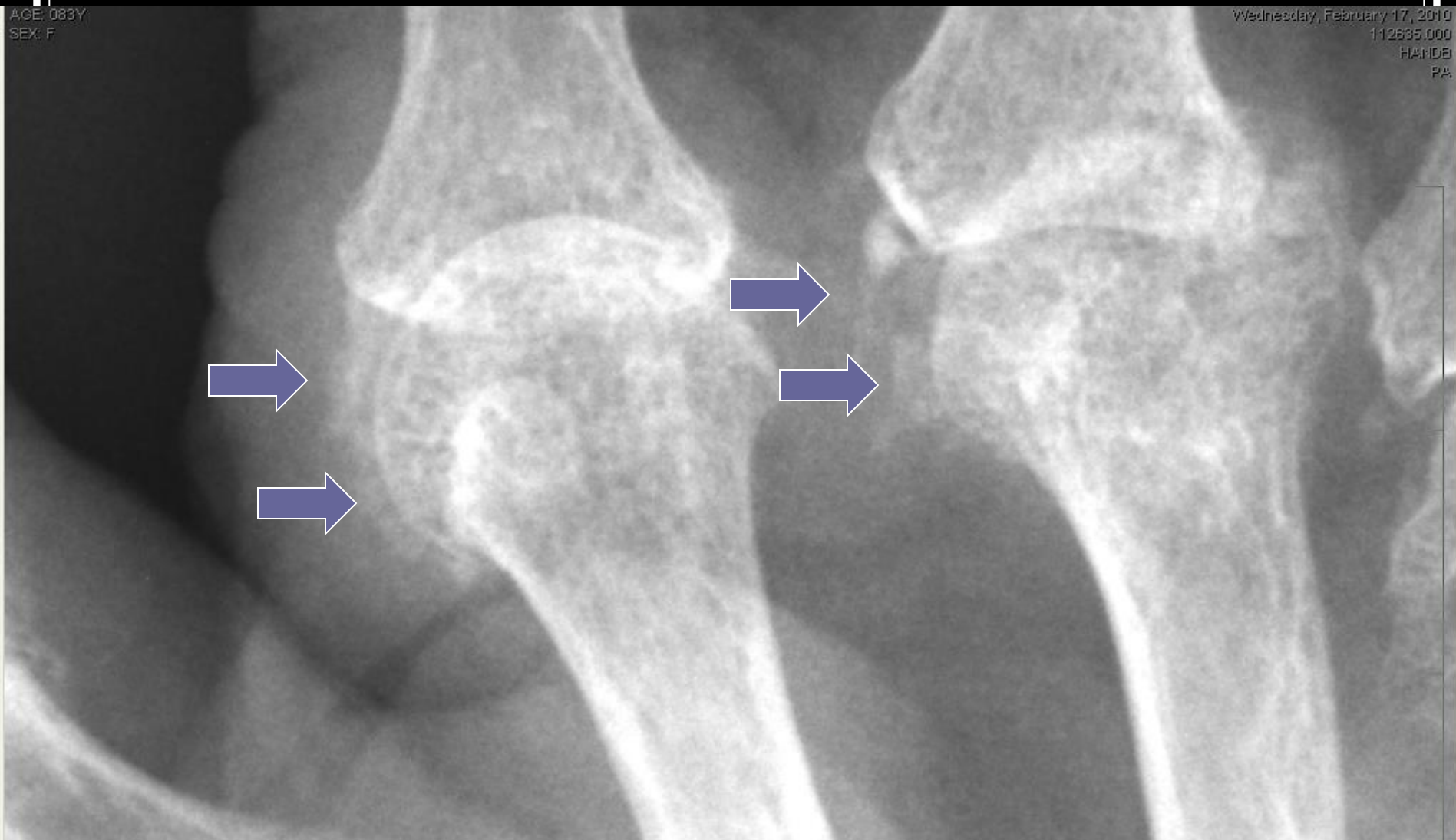
AGE: 083Y
SEX: F

Wednesday, February 17, 2010
112635.000
HAND
PA



AGE: 083Y
SEX: F

Wednesday, February 17, 2010
112535.000
HANDS
PA



AGE: 078Y
SEX: F

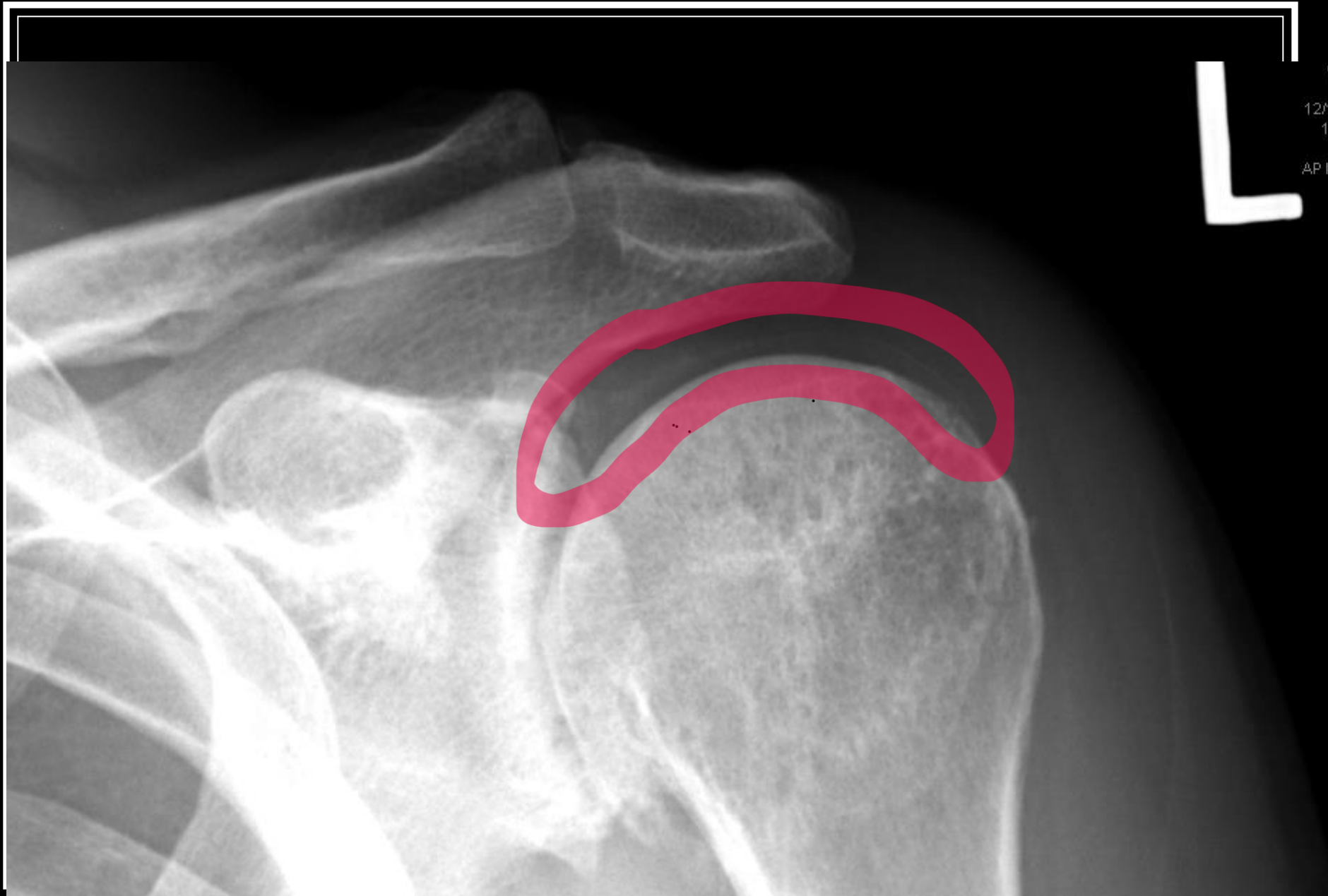
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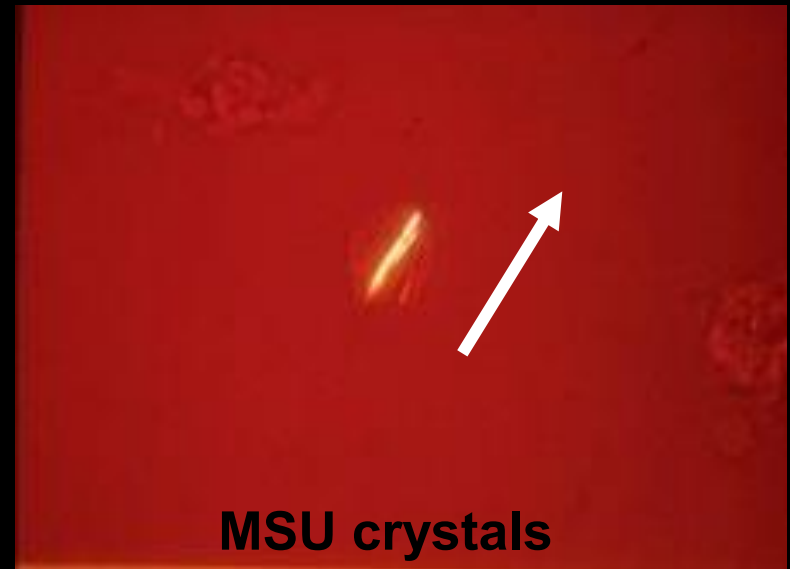
Ref. Phys: LEE, STEVE

Plate ID: 91042857
Relative Exposure: 20

Metacarpophalangeal osteophytosis and chondrocalcinosis



CPPD Crystals



- CPPD crystals are weakly positively birefringent, rhomboid, rods, squares, or irregular under compensated polarized light

1. <http://www.emedicine.com/med/topic1938.htm>.

2. ACR Clinical Slide Collection on the Rheumatic Diseases, 1998

Risk Factors for CPPD

SPORADIC

Advanced age

Prior joint trauma

Hemochromatosis

Hyperparathyroidism

Hypomagnesemia

Gout

Medications

Hypophosphatasia

Hypothyroidism

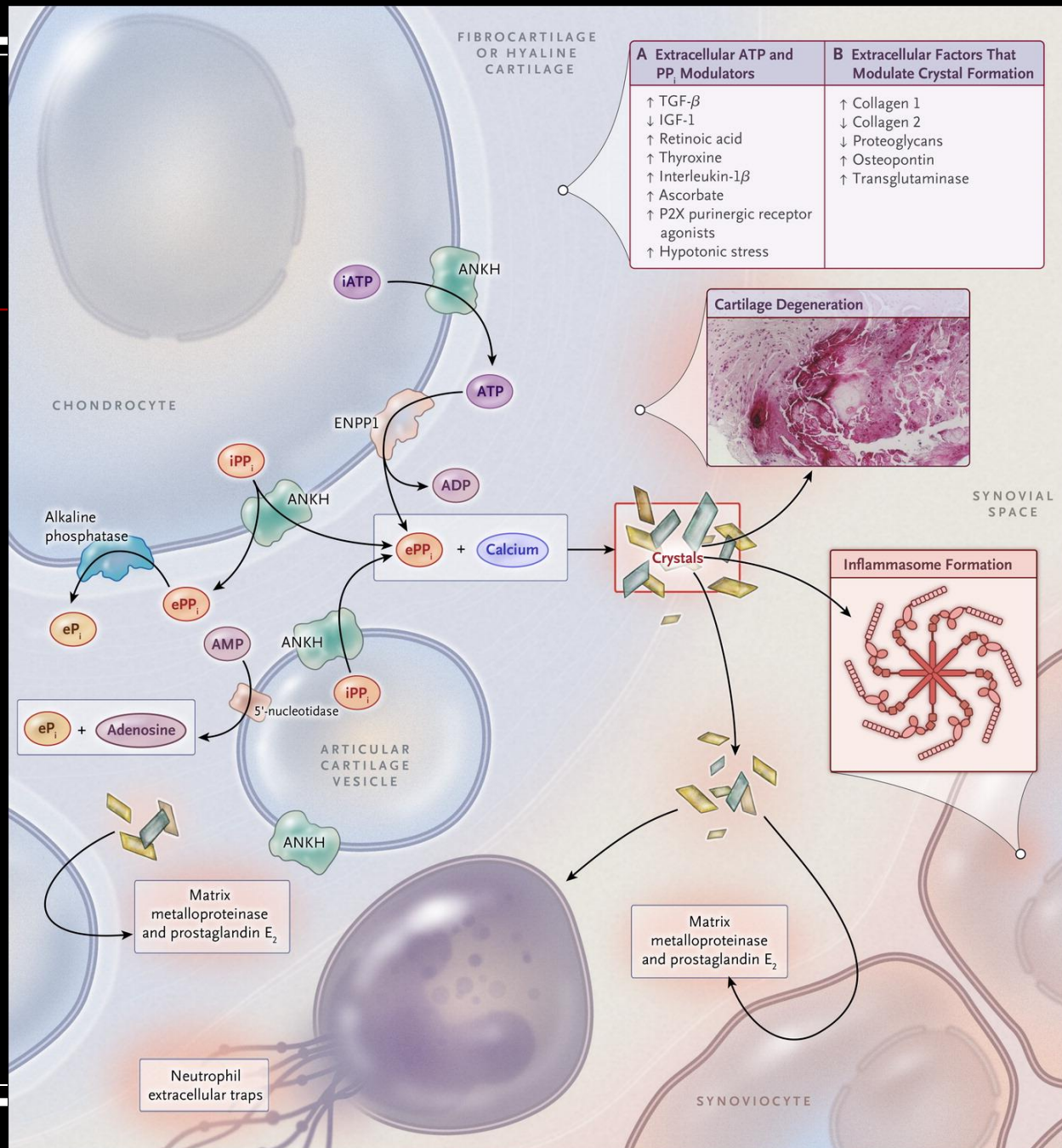
FAMILIAL

ANK mutations

Other mutations

ANK mutations

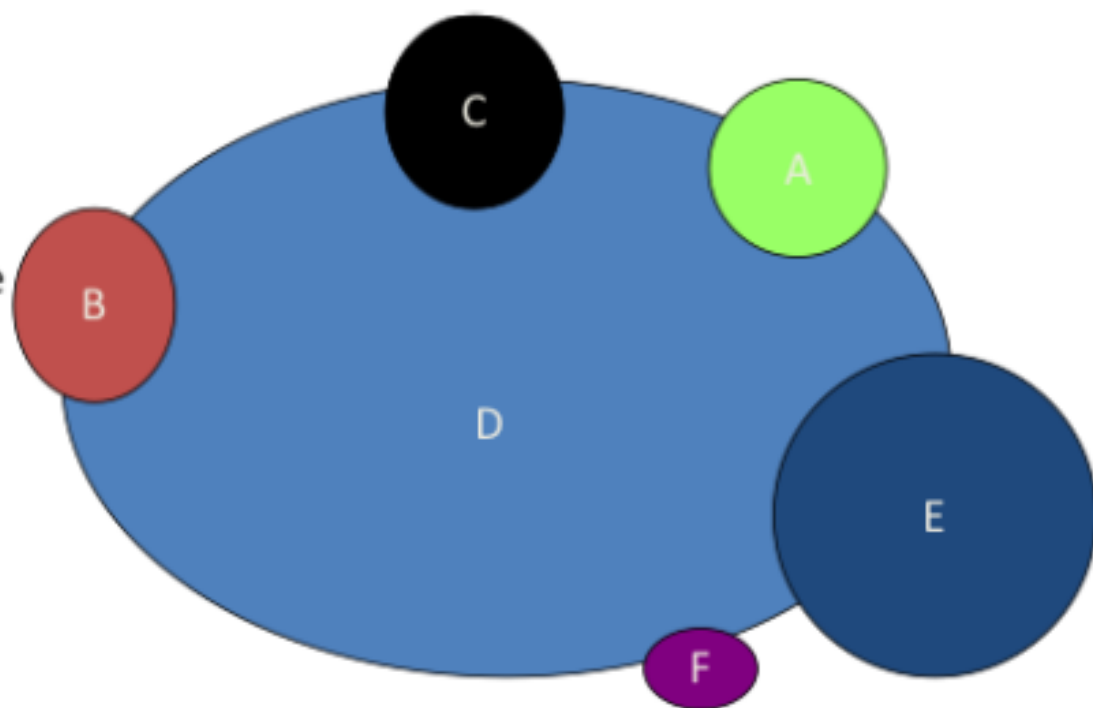
- Transmembrane protein transporter of inorganic pyrophosphate (anionic intracellular component of cppd)
- Intracellular ATP degradation $\rightarrow$ P_{Pi}
- High levels of ANK in cppd joint tissues





Relative frequencies of different presentations of CPPD arthritis

- A. Acute CPPD (pseudogout)
- B. Rheumatoid-like
- C. OA-like with acute attacks
- D. OA-like without acute attacks
- E. Asymptomatic
- F. Neuropathic-like



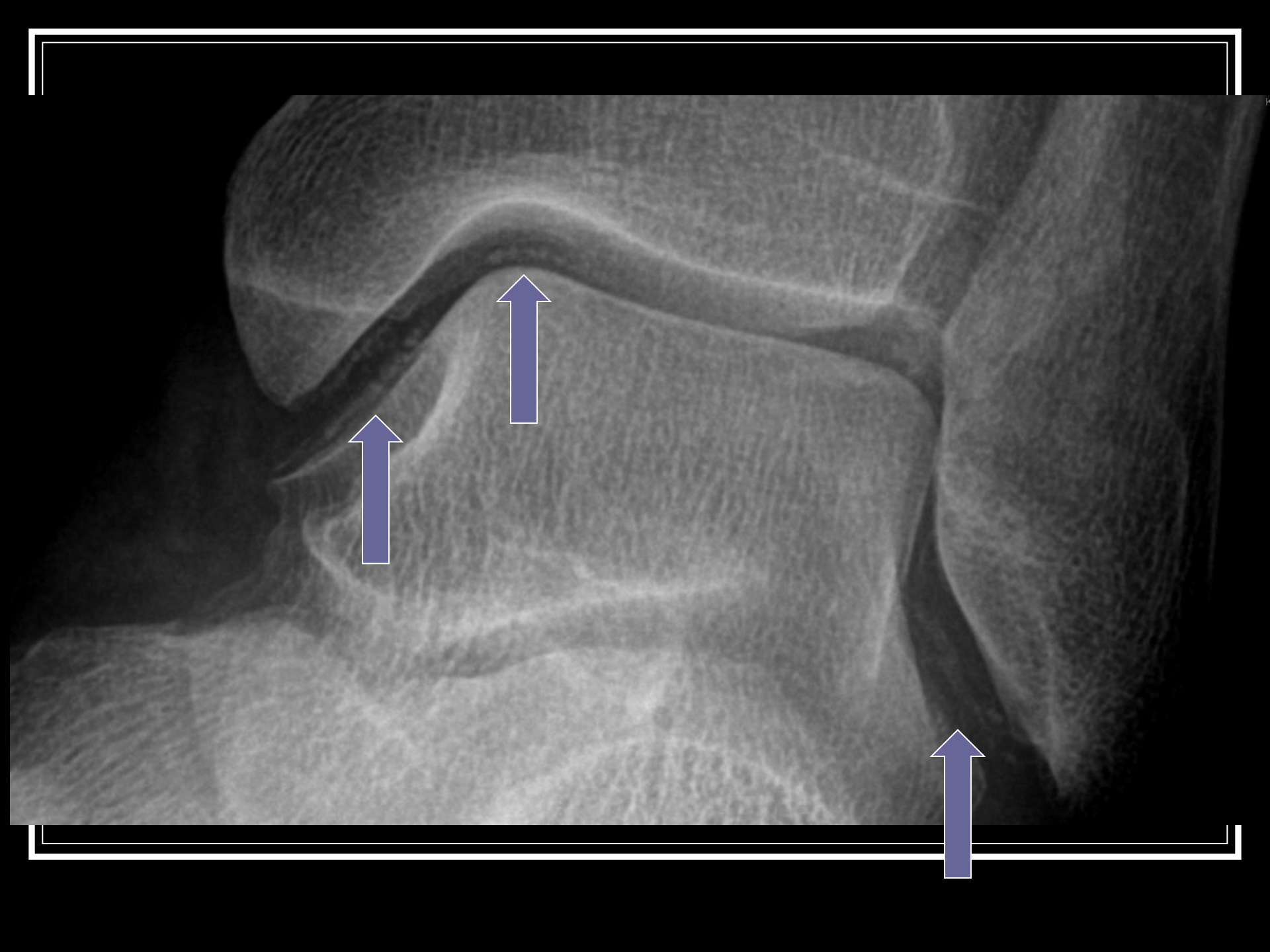
Acute CPPD vs. gout

	CPPD	Gout
Joint involvement	Knee/wrist	Toe/ ankle
Synovial fluid	Bloody	Not bloody
Length of attack	Days-weeks	Days
Lightening attacks	Absent	Present









AGE: 061 Y
SEX: M

1/11/20
12:15:
HAND
A



Ref. Phys: LEE
ACC: 15579028

Fi
Plate II
Exhibits Exposed - 16

Triangular fibrocartilage chondrocalcinosis

AGE: 0627
SEX: F

Thursday, October 15, 2009
11:31:46.000
HANDB
AP

Ref. Phys: LEE

HFS
Plate ID:
Relative Exposure: 2044



AGE 085Y
SEX M



Ref: Provoc. OTSUWUJOHN, TSUYOSHI

Plate 1
Relative E.





New and emerging therapies

Drug

- Probenecid
- Phosphocitrate
- Drugs which augment the action of alkaline phosphatase

Potential action

- Interferes with ANK-mediated ATP release
- Prevents crystal-cell interactions
- Augment crystal dissolution

EULAR CPPD Treatment Guidelines

- Ice, arthrocentesis and steroid injection
- NSAIDs as tolerated
- Low dose colchicine up to QID acutely
- Systemic corticosteroids (intramuscular, PO)
- Long term:
 - Low dose colchicine qd to bid
 - Methotrexate
 - Hydroxychloroquine
- Correct underlying risk factors:
 - CKD and hyper PTH and phosphate disorders

In Summary:

- Common cause of mono- or poly-arthritis
- Calcium pyrophosphate crystals
- 3% of >60 yo; 50% of >90 yo
- Other risk factors
- Knees, wrists, mcp's
- Conservative and pharmacologic treatments
 - Ice, rest
 - Colchicine
 - Steroids

Questions about CPPD?

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